

MICHAEL CLOUGHER

Software engineer with **5+ years** building and operating a production Python platform that orchestrates **tens of millions of daily marketing sends** across **~285 brand domains and accounts**. Sole developer on a **75K+ line codebase** spanning campaign execution, an in-house HTML rendering engine, statistical content selection, Cloud Run services, and the CI/CD that ships it all. Accounts on the automated platform see **10–100% revenue lift** vs. manual scheduling. M.S. Computer Science, Johns Hopkins (GPA 3.96).

EXPERIENCE

Software Engineer · RXMG (Remote · Irvine, CA)

Aug 2024 – Present

Internal title: Digital Marketing Analyst, Team Lead

- Architect and maintain **Onix**, a multi-ESP campaign execution engine handling scheduling, template publishing, segmentation, and delivery tracking for tens of millions of daily events.
- Designed a **clean domain abstraction over two structurally different ESPs** (Iterable's campaign+template separation vs. Pinpointe's combined newsletter model) so the core logic stays ESP-agnostic and adding a third ESP would be an adapter, not a rewrite.
- Built and shipped an in-house **HTML rendering engine** (Jinja-based brand shells, dark-mode treatments, dynamic legal-text resolution, copyright-year refresh) replacing a legacy third-party publish API; cut over dozens of brand domains in waves, shipping without major production incidents.
- Designed an **ESP-specific A/B test** on tracking-link rewrites; ran the eCPM audit informing the per-ESP keep/revert decision.
- Built an **editor-side validation system** (Apps Script in Google Sheets, backed by a self-hosted LanguageTool service) that gates publish on merge-tag, encoding, footnote, link, year-literal, duplicate-line, spelling, and grammar checks — moving content QA upstream to where the content team actually works.
- Built **suppression infrastructure**: per-offer MD5 cross-referencing at drop creation, a global-suppression API endpoint for internal uploads, and a pair of Cloud Run jobs (one per ESP) that continuously sweep segments and unsubscribe matches.
- Stood up **CI/CD from scratch**: GitHub Actions, **1,000+ pytest cases across 48 test files** covering rendering, CTA detection, scheduling, and merge-tag validation; Docker images deployed to Google Cloud Run via a service-selector workflow.
- Built **concurrent ESP integrations** with retry and exponential backoff; separate Selenium-based automation for a legacy ESP that lacks an API.

TECHNICAL SKILLS

Languages

Python, SQL, JavaScript (Google Apps Script), Bash, Jinja2, HTML/CSS

Backend & Data

pandas, NumPy, SQLAlchemy, MySQL, BigQuery, dbt, beautifulsoup4, Flask

Cloud & Infrastructure

Google Cloud Platform (Cloud Run, GCS, Cloud SQL, Secret Manager, Artifact Registry, VPC/NAT), Docker, GitHub Actions CI/CD

Statistics & Applied Modeling

Rolling-window aggregations, multi-level groupby analysis, tiered ranking, scikit-learn preprocessing; coursework in regression, NLP, and machine learning (Johns Hopkins MS CS)

Testing & Quality

pytest (1,000+ cases), diff harnesses for risky migrations, gated canary rollouts, structured logging + Slack alerting

Other Tools & Integrations

Selenium, Slack SDK, OpenAI API, Google Sheets (pygsheets) + Apps Script, Smartsheet, SFTP, Tableau; integrations with Iterable, Pinpointe, and other marketing platforms

EDUCATION

Johns Hopkins University

M.S. Computer Science · GPA 3.96 · Jan 2019 – May 2021

Concentrations in Data Science and Bioinformatics

Connecticut College

B.A. · GPA 3.65 · Aug 2011 – May 2015

Triple major: Computer Science, Mathematics (Statistics concentration), Economics

Software Engineer · RXMG (Remote)

Dec 2023 – Aug 2024

Internal title: Senior Digital Marketing Analyst

- Expanded **Viper**, the Python scheduling platform, into the production system managing content selection across the full ~285-account portfolio.
- Designed a **statistical content-ranking engine** — rolling-window aggregations, multi-level groupby analysis, tiered scoring — to select email creative per account and domain.
- Integrated the platform with Google Sheets, MySQL, and internal APIs into a fully automated pipeline from campaign planning through execution.
- **Accounts on the automated platform saw 10–100% increases in monthly revenue** vs. manual scheduling.

Software Engineer · RXMG (Remote)

May 2021 – Dec 2023

Internal title: Digital Marketing Analyst II / Digital Marketing Analyst

- **Built Viper from scratch** — a Python scheduling system that replaced thousands of hours/month of manual campaign setup across the team.
- Created Tableau dashboards translating campaign and delivery data into actionable views for technical and non-technical stakeholders.

Data Analyst · Rocket (San Francisco, CA · recruiting-tech)

Sep 2019 – Nov 2020

- Python-based exploratory analysis of sourcing and acquisition system performance; ad-hoc analyses and scripts for product and operations teams.

Computer-Aided Design Engineer · ODMS (San Leandro, CA)

Mar 2016 – Aug 2019

- Designed parametric CAD models in Autodesk Fusion 360 for rapid prototyping and small-scale manufacturing; hands-on with FDM and SLA 3D printing, metal-casting design (sprues, gating, shrinkage tolerances).

HONORS

- **U.S. National Rowing Team** — three World Championships (2017, 2018, 2022)
- Brown/Brooks Award — scholarship, leadership, sportsmanship
- Ward S. Donner Award — athletics, studies, character